IAN SNIDER

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Roboticist designing autonomous aerial vehicles for radiation detection.

EDUCATION

Ph.D. Nuclear Engineering - Robotics - University of California, Berkeley — *Aug 2025 - present*

B.S. Mechanical Engineering - Washington University in St. Louis — 4.00/4.00 — *May 2025*

B.A. Physics, Math minor - Truman State University — 3.89/4.00 — *Dec 2024*

EXPERIENCE

Lawrence Berkeley National Laboratory - *Lab Affiliate*, *Berkeley, CA* — *Aug 2025 - present*

- Developed a method for a hexacopter to control the trajectory of a 10m tether-suspended radiation detector
- Designed and implemented safety-critical software infrastructure for autonomous systems and outer-loop control algorithms for commanding ArduPilot inner-loop controller
- Architected a communication pipeline for GCS, transmitter, companion computer, and Cube Blue flight controller
- Executed autonomous flight tests at LBNL and Richmond Field Stations under Part 107

Lawrence Livermore National Laboratory - *Graduate Intern*, *Livermore, CA* — *June 2025 - Dec 2025*

- Developed CRISP (CRItical Simulation Pipeline), a Python tool for distributing high-performance computing (HPC) jobs
- Used 2 world-class LLNL HPC clusters Dane and Ruby with Slurm sbatch/srun scripts
- CRISP successfully automated compilation and benchmarking workflows for ~3420 reactor simulation cases

Brookhaven National Laboratory - *Student Researcher*, *Upton, NY* — *Summers 2022 - 2024*

- Developed BRR (Bayesian Resonance Reclassifier) for reclassification tasks on HPC clusters
- Applied machine learning using Scikit-learn to classify nuclear data
- Used the transport code OpenMC and HPC clusters to parallel process sensitivity analyses for reactor models

Truman State University - *Student Astronomy Researcher*, *Kirksville, MO* — *2021 - 2022*

- Calculated trajectories for ~3000 Starlink satellites to optimize telescope viewing plans and developed a GUI

PROJECTS

Sniff Consensus Protocol for Noisy Multi-Agent Formations - *Developer* — *Fall 2025*

- Consensus protocol for a random graph of agents perturbed with communication noise
- Used Erdos-Renyi graphs to generate adjacency matrices and applied communication noise with Random Matrix Theory

WashU Robotics MateROV & MARINER Projects - *Mechanical Lead* — *2023 - 2025*

- Designed and tested an autonomous underwater vehicle (AUV)
- Prototyped dive control using an IMU and depth state estimation with a Kalman Filter for 1D hovering
- Used Arduino for programming AUV sensors and control system
- Built a chassis and buoyancy engine

CrazyFlie 2.1 cascaded PID/PD Control System - *ESE 4481 Student* — *Spring 2025*

- Designed a control system for a CrazyFlie 2.1 quadcopter and achieved stable hovering
- Used Simulink and MATLAB for simulating the control system response and finding stability margins
- Used C to program the quadcopter embedded system

SKILLS

- **Programming:** C, C++, Python, Go, Bash, Lua, MATLAB, Mathematica, Octave, LaTeX, Typst, Vimscript
- **Software/Technical:** MAVLink, ROS2, Simulink, SolidWorks, OpenMC, Scikit-learn, Slurm, Git, Linux
- **Physics/engineering:** Autonomous Aerial Vehicle Control, Classical Mechanics, Electrodynamics, Vibrations, Mathematical Physics, Fluid Mechanics, Solid Mechanics, Acoustics, Materials Science, Thermal Systems, Propulsion
- **Math:** Linear Algebra, ODEs, Computing Structures, Control Systems, Machine Learning, Random Walks, Random Matrix Theory, and Optimizations